Uniform prefix growth in an internal pair equality

Human Compression project: L044 continuation

Retain the complete hypotheses of Theorem 4 of *Ambient transport and internal pair deficits*, including $k \notin F$, intransitivity of the full local $K = G_{>k}$ on the actual $H = G_{\geq k}$-orbit O, and equality $|F| = q - 1$ with $|O| = 2q$, $q \geq 3$. Thus $K|_O$ is the diagonal S_q on two natural rows, and every tail mark outside $F \cup \{k\}$ fixes O. Let $a = \min F$ and $x = \max F = a + q - 2$; here a denotes a label, not the crossing vertex used elsewhere. Put $P = G_{<k}$. Suppose also that a specified set $C \subseteq \mathrm{Fix}(K)$ has size $h \geq 1$. The ambient group G is transitive.

Lemma 1 (A third fixed row)**.** *Every point of the union $O^P = \bigcup_{p \in P} O^p$ is fixed by $G_{>x}$. This union is a disjoint union of K-orbits of size q, is disjoint from C, and has at least $3q$ points. Consequently*

$$|\mathrm{Fix}(G_{>x})| \geq 3q + h. \tag{1}$$

Proof. The two rows of O are actual K-orbits of size q. The prefix P centralizes K, so each of their translates is again a K-orbit of size q. Distinct orbits are disjoint and none contains a point of C, all of whose points are fixed by K. If $O^P = O$, then P preserves O; so does H, by its definition. Since $G = \langle P, H \rangle$, transitivity would give $O = \Omega$. This is impossible because C is nonempty and disjoint from O. Thus at least one further q-point K-orbit occurs in O^P.

Every mark after x lies outside $F \cup \{k\}$ and hence fixes O pointwise. Also $G_{>x} \leq K$, so it commutes with P. For $v \in O$, $p \in P$ and $g \in G_{>x}$, one has $(v^p)^g = (v^g)^p = v^p$. Thus $G_{>x}$ fixes O^P pointwise. It fixes C as a subgroup of K. The disjointness already proved gives (1). This argument does not require K to be transitive on any larger set or identify two distinct row orbits. □

Lemma 2 (Every intervening prefix adds a row)**.** *Under the same hypotheses,*

$$|\mathrm{Fix}(G_{>x})| \geq (a - k + 2)q + h. \tag{2}$$

Proof. Put $Q = \langle \rho_j : j \in F \rangle$ and $Z_j = O^{G_{\leq j}}$ for $k - 1 \leq j \leq a - 2$. The two natural K-rows of O are also natural Q-orbits of size q. Each prefix $G_{\leq j}$ in this range centralizes Q, so Z_j is a union of q-point Q-orbits. It excludes the nonempty Q-fixed set C, hence is a proper subset of Ω.

For $k - 1 \leq j \leq a - 3$, any mark of index at least $j + 2$ outside F fixes O: its index is at least $k + 1$, so it lies outside $F \cup \{k\}$. It commutes with $G_{\leq j}$ and therefore fixes Z_j pointwise. The marks in F preserve Z_j, and so does the prefix $G_{\leq j}$. Consequently ρ_{j+1} cannot preserve Z_j, or every generator of G would preserve this proper set. Because $j + 1 \leq a - 2$, it commutes with Q and any new point brings a whole additional q-point Q-orbit. Thus $|Z_{j+1}| \geq |Z_j| + q$.

Lemma 1 starts this chain with $|Z_{k-1}| \geq 3q$. There are $a - k - 1$ steps, including none when $a = k + 1$. Hence $|Z_{a-2}| \geq (a - k + 2)q$. Every mark after x fixes O and commutes with $G_{\leq a-2}$, so fixes Z_{a-2} pointwise. This set is disjoint from the K-fixed set C, which the later tail also fixes. This proves (2). □

Corollary 3 (The uniform source exit)**.** *Put $X = \Omega \setminus \mathrm{Fix}(G_{>x})$ and $n = |\Omega|$. In either source head configuration $(k, h) = (2, 2)$ or $(5, 7)$, every starting label $a \geq k + 1$ gives*

$$x \leq \frac{n - |X| - 1}{2}. \tag{3}$$

This is alternative (a) of source Proposition 5.31. If also $x = r - 1$ and H fixes C pointwise, then $r \le (n-1)/2$.

Proof. Write $a = k + 1 + d$ with $d \ge 0$. Since $x = a + q - 2$, the difference between the lower bound (2) and $2x + 1$ is

$$d(q-2) + q + h - 2k + 1.$$

This is $d(q-2) + q - 1 \ge 2$ for $(k, h) = (2, 2)$, and $d(q-2) + q - 2 \ge 1$ for $(k, h) = (5, 7)$. Thus (3) follows.

For the last assertion, the initial union $Z_{k-1} = O^P$ consists of nontrivial K-orbits and is disjoint from $\mathrm{Fix}(K)$. The whole fixed set of K is preserved by P. If it were exactly C, the additional hypothesis that H fixes C would make C a proper nonempty G-invariant set, a contradiction. Therefore there is a K-fixed point outside C. All Z_j consist of nontrivial Q-orbits, so this additional point is outside Z_{a-2} as well. When $x = r - 1$, every point is fixed by $G_{>x}$; hence $n \ge (a - k + 2)q + h + 1 \ge 2x + 3$ in both head configurations. This is $r = x + 1 \le (n-1)/2$. □

Source and scope. The target is Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1, Proposition 5.31(a) and its internal-orbit cases, physical pages 38 and 42–43. Its two head configurations supply two or seven specified points fixed by H, and hence by K; an application must identify these actual points. The fixed-support target is borrowed. The third-row and prefix-growth arguments cover every path start while retaining all three flip patterns. The complete internal-pair proof and its ambient-eligibility hypotheses remain paid. This is an additive strengthening of the frozen one-page row exit; its proof of Lemma 1 is unchanged. Other deficits, source tables and full L044 remain pending; no computation, formal theorem or question credit is added.